

PROBABILITY THEORY

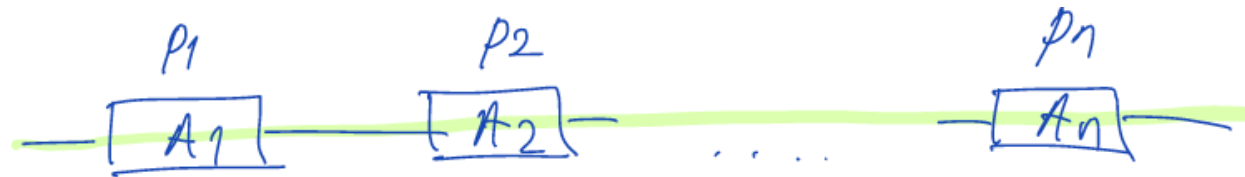
Session 5

Solving real life problems

Topics:

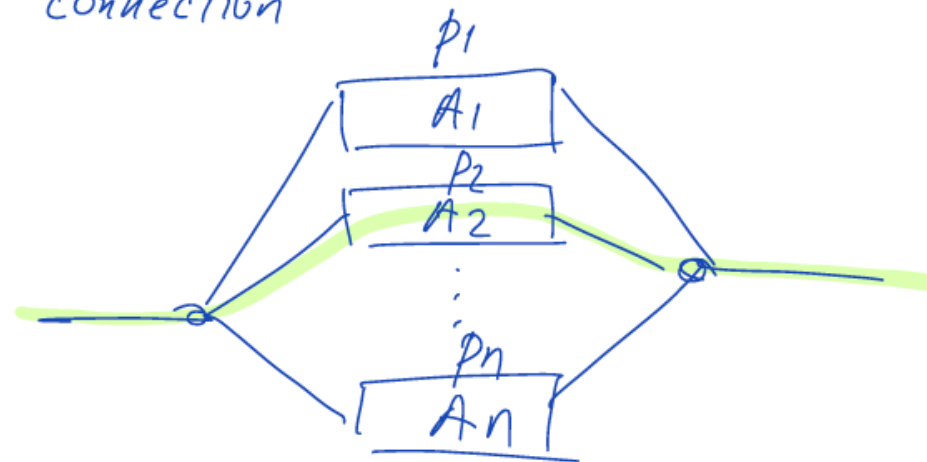
- Serial and parallel connections
- Birthday paradox
- Moment generation function
- Multi-dimensional r.vs.
- Many examples :)

Serial and parallel connection



all components are working

Parallel connection

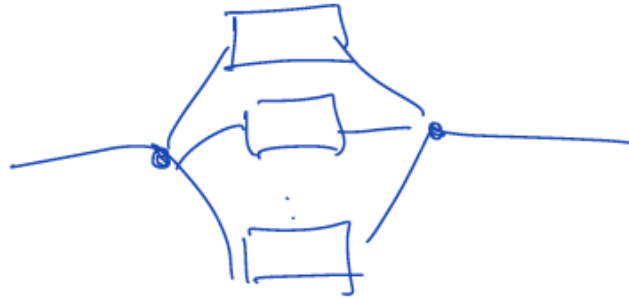


at least one of the components is working

Serial connection

$$\begin{aligned} P(\text{system works}) &= P(\text{all comp. work}) = \\ &= P(\underbrace{A_1 \text{ works}}_{\text{event } A_1} \overset{\text{AND}}{\cap} \underbrace{A_2 \text{ works}}_{\text{event } A_2} \cap \dots \cap \underbrace{A_n \text{ works}}_{\text{event } A_n}) = \uparrow A_1 \dots A_n \text{ are independent} \\ &= P(\text{event } A_1) \cdot P(\text{event } A_2) \cdot \dots \cdot P(\text{event } A_n) = \\ &= \boxed{p_1 \cdot p_2 \dots \cdot p_n} \end{aligned}$$

Parallel connection



$$Pr(\text{system works}) = P(\text{at least one of the comp. works}) =$$

Y - # of working comp.

$$P(\text{exactly one comp.}) = P(Y=1)$$

$$P(\text{at least one}) = P(Y \geq 1) =$$

$$= P(Y=1) + P(Y=2) + \dots + P(Y=n) =$$

n terms

$$= 1 - P(Y < 1) = 1 - P(Y=0)$$

$$P(\text{at least two}) = P(Y \geq 2) =$$

$$= 1 - P(Y < 2) = 1 - P(Y=0) - P(Y=1)$$

$$\Pr(\text{system works}) =$$

$$= 1 - P(\text{zero comp. are working}) =$$

$$= 1 - P(A_1 \text{ does not work}, \dots, A_n \text{ does not work}) =$$

$$= 1 - \underbrace{P(\bar{A}_1)}_{1-p_1} \cdot \dots \cdot \underbrace{P(\bar{A}_n)}_{1-p_n} = 1 - (1-p_1) \dots (1-p_n)$$

ONLY
if independent



Birthday paradox

If k people are at a party, what is the probability that at least two of them have the same birthday? Suppose that there are $n=365$ days in a year and all days are equally likely to be the birthday of a specific person.

Birthday paradox

If k people are at a party, what is the probability that at least two of them have the same birthday? Suppose that there are $n=365$ days in a year and all days are equally likely to be the birthday of a specific person.

$$\begin{array}{ll} \text{when } k=23 & p \approx 0.50 \\ k=57 & p \approx 0.99 \end{array}$$

Let A be an event = at least 2 persons have birthday on the same day
it is easier to $\uparrow$ work with A^c = nobody has the same birthday

$$P(A) = 1 - P(A^c) = 1 - \frac{|A^c|}{|S|}$$

$|S|$: total number of possible outcomes. We consider sequence of birthdays of k persons; ordered; with replacement
 $\Rightarrow |S| = n^k = 365^k$

$|A^c|$: We consider sequence of birthdays, but they can not be the same $\Rightarrow$ ordered, without replacement $\Rightarrow$
 $|A^c| = n \cdot (n-1) \cdot \dots \cdot (n-k+1) = 365 \cdot 364 \cdot \dots \cdot (365-k)$

Independent events

- We roll a dice twice. Let us define A as the event that the first outcome is odd. Let E be the event that the second outcome is 6. Let B be the event that both outcomes are the same. Finally, let C be the event that the sum of outcomes is even.
- Are A and E independent?
- Are A and B independent?
- Are B and C independent?

$$A = \{1, 3, 5\}$$

of outcomes in A

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

of outcomes in S

$$B = \{(1,1), (2,2), \dots, (6,6)\}$$

of outcomes in B

$$P(B) = \frac{6}{36} = \frac{1}{6}$$

total # of outcomes

$$A \cap B = \{(1,1), (3,3), (5,5)\}$$

$$P(A \cap B) = \frac{3}{36} = \frac{1}{12}$$

We check definition for independency:

$$\frac{1}{12} = P(AB) = P(A)P(B) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$$

$\Rightarrow$ A and B are independent

Now we introduce C:

$$C = \left\{ \begin{array}{l} (1,1), (1,3), (1,5) \\ (2,2), (2,4), (2,6) \\ (3,1), (3,3), (3,5) \\ (4,2), (4,4), (4,6) \\ (5,1), (5,3), (5,5) \\ (6,2), (6,4), (6,6) \end{array} \right\}$$

There are
18 elements in C

$$P(C) = \frac{18}{36} = \frac{1}{2}$$

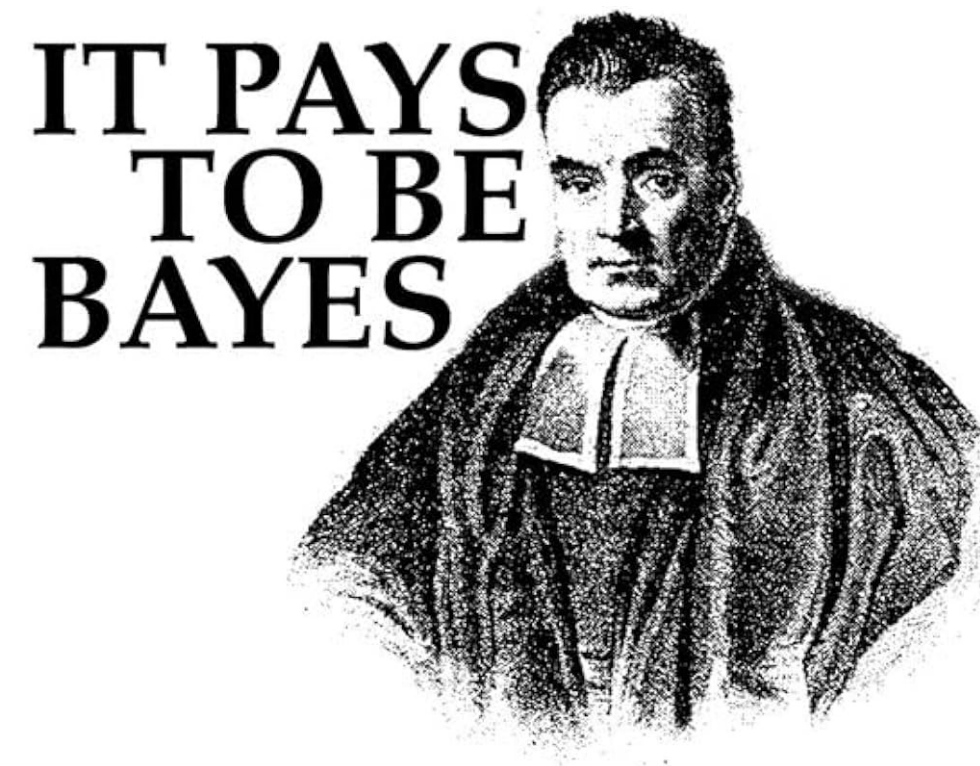
$$BC = \{ (1,1), (2,2), \dots, (6,6) \}$$

$$P(BC) = P(B) = \frac{1}{6}$$

we check whether definition of independency holds:

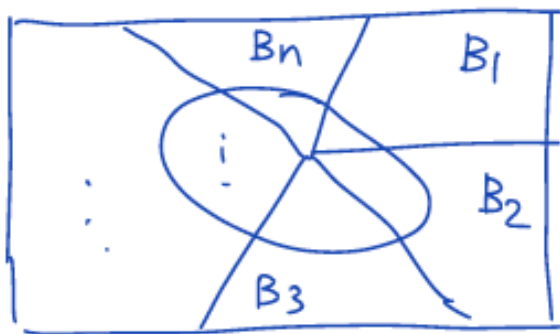
$$\frac{1}{6} = P(BC) \neq P(B) \cdot P(C) = \frac{1}{6} \cdot \frac{1}{2}$$

$\Rightarrow$ B and C are not independent



Bayes' formula

Formula can be generalized to a case of n subsets: $B_1, B_2, \dots, B_n$



1) $B_1, \dots, B_n$ are mutually exclusive

$$2) AB_1 \cup AB_2 \cup \dots \cup AB_n = A$$

$$\begin{aligned} P(A) &= P(AB_1) + \dots + P(AB_n) = \\ &= P(A|B_1)P(B_1) + \dots + P(A|B_n) \cdot P(B_n) \end{aligned}$$

formula of total probability

Bayes formula will look like this:

$$P(B_1 | A) = \frac{P(A|B_1)P(B_1)}{P(A|B_1) \cdot P(B_1) + \dots + P(A|B_n) \cdot P(B_n)}$$

we can write similar formulas for all $B_i, i=1 \dots n$

Example

- On a road during a month there are driving 2.000 scooters, 4.000 cars and 6.000 trucks. Probability that a scooter is having an accident is 0.01; for a car it is 0.03 and for a truck it is 0.015.
- There is an accident. What is the probability that a scooter is involved in the accident?

Questions

Suppose X has mean 2 and variance 3

- Compute $\text{Var}(3X)$
 - 3
 - 9
 - 27
- Compute $\text{Var}(3X+4)$
 - 27
 - 31
 - 43
- Compute $E[X^2]$
 - 4
 - 7
- Compute $\text{Var}(X^2)$
 - 9
 - Can not be computed from information given

Example

Let X be a noise voltage that is uniformly distributed in $S = \{-3, -1, +1, +3\}$ with $P(X = k) = 1/4$.

- (a) Find $E[Z]$ where $Z = X^2$.
- (b) The noise voltage X is amplified and shifted to obtain $Y = 2X + 10$, and then squared to produce $Z = Y^2$. Find $E[Z]$.

The nth moment of r.v. X

- The mean value is sometimes called the first moment of X .
- **Definition.** The **nth moment** of X is defined by

$$E[X^n] = \sum_i x_i^n p(x_i)$$

$$E[X^n] = \int_{-\infty}^{\infty} x^n f(x) dx$$

Moment generating function

- **Definition.** The **moment generating function** of a r.v. X is defined for all values t by

$$\varphi(t) = E[e^{tX}] = \sum_i e^{tx_i} p(x_i)$$

$$\varphi(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

- This function is called moment generating function, because all moments of X can be obtained by successively differentiating it.
- There is a one-to-one correspondence between the moment generating function and the distribution function of a r.v.: the mgf uniquely determines the distribution and vice versa.

How to obtain $E[X]$ and $\text{Var}(X)$ from moment generating function?

$$\varphi'(t) = \frac{d}{dt} E[e^{tX}] = E\left[\frac{d}{dt}(e^{tX})\right] =$$

$$= E[X e^{tX}]$$

$$\varphi'(0) = E[X]$$

$$\varphi''(t) = \frac{d}{dt} E[X e^{tX}] = E\left[\frac{d}{dt}(X e^{tX})\right] =$$

$$= E[X^2 e^{tX}]$$

$$\varphi''(0) = E[X^2]$$

$$\text{Var}(X) = \varphi''(0) - \varphi'(0)^2$$

MGF for Poisson r.v.

Moment generation function:

$$\begin{aligned}\varphi(t) &= E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} e^{-\lambda} \frac{\lambda^i}{i!} = e^{-\lambda} \sum \frac{(\lambda e^t)^i}{i!} = \\ &= e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}\end{aligned}$$

$$\varphi'(t) = \lambda e^t e^{\lambda(e^t - 1)} \quad \varphi'(0) = \lambda \cdot 1 \cdot e^{\lambda(1-1)} = \lambda$$

$$\Rightarrow E[X] = \lambda$$

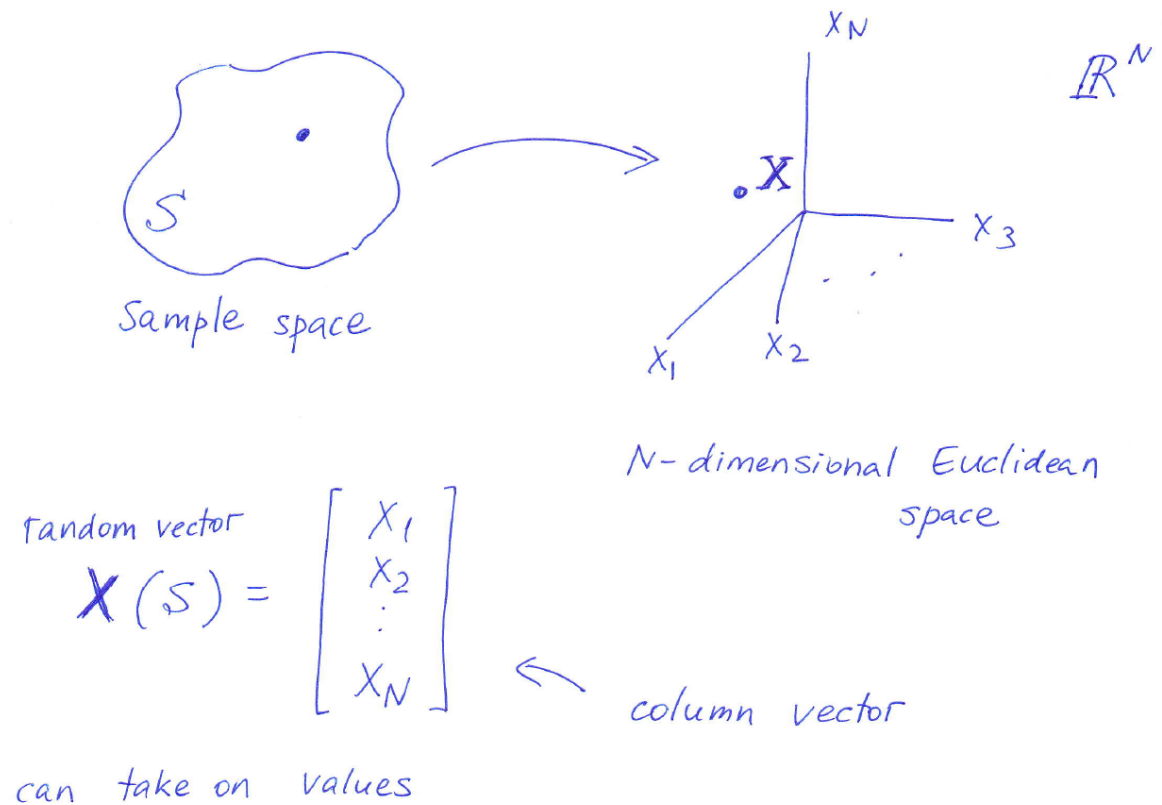
$$\varphi''(t) = (\lambda e^t)^2 e^{\lambda(e^t - 1)} + \lambda e^t e^{\lambda(e^t - 1)}$$

$$\Rightarrow E[X^2] = \varphi''(0) = \lambda^2 + \lambda$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Multidimensional r.vs. = random vectors

- A random vector is a mapping from the sample space S to a numerical sample space, N -dimensional Euclidean space.



$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$$

Discrete case: joint probability mass function

$$P_{X_1, X_2, \dots, X_N}[x_1, x_2, \dots, x_N] = P[X_1 = x_1, \dots, X_N = x_N]$$

vector notation :

$$P_{\mathbf{X}}[\mathbf{x}] = P[\mathbf{X} = \mathbf{x}]$$

$$\text{cdf: } F_{X_1, \dots, X_N}(x_1, \dots, x_N) = P[X_1 \leq x_1, \dots, X_N \leq x_N]$$

Marginal pmf vs joint pmf

$$P[X_1 = x_1] = P[X_1 = x_1, X_2 \in S_{X_2}, X_3 \in S_{X_3}, \dots]$$

$$p_{X_1}[x_1] = \sum_{\text{all } x_2, \dots, x_N} \dots \sum p_{X_1, \dots, X_N}[x_1, \dots, x_N]$$

- The marginal pmfs do not determine the joint pmf, unless the random variables are independent.
- Random variables are defined to be independent if the joint pmf factors:

$$p_{X_1, \dots, X_N}[x_1, \dots, x_N] = p_{X_1}[x_1] \cdot p_{X_2}[x_2] \cdot \dots \cdot p_{X_N}[x_N]$$

Expected values

Def.

$$E_X[X] = \begin{bmatrix} E_{X_1}[X_1] \\ E_{X_2}[X_2] \\ \vdots \\ E_{X_N}[X_N] \end{bmatrix}$$

- Expectation of a scalar function of X?

$$g(X_1, X_2, \dots, X_N)$$

$$E_{X_1, \dots, X_N} [g(X_1, \dots, X_N)] =$$

$$= \sum_{x_1} \dots \sum_{x_N} g(x_1, \dots, x_N) p_{X_1, \dots, X_N} [x_1, x_2, \dots, x_N]$$

Linearity of expectation

- Expectation of a sum

$$E_{X_1, X_2, \dots, X_N} \left[\sum_{i=1}^N X_i \right] =$$

$$= \sum_{x_1} \dots \sum_{x_N} (x_1 + x_2 + \dots + x_N) p_{X_1, \dots, X_N} [x_1, \dots, x_N] =$$

$$= \sum \dots \sum x_1 p_{X_1, X_2, \dots, X_N} [x_1, \dots, x_N] + \dots$$

$$+ \sum \dots \sum x_N p_{X_1, X_2, \dots, X_N} [x_1, \dots, x_N] =$$

$$= E[X_1] + \dots + E[X_N]$$

Linearity of expectation

- By a slight modification we can show that

$$E_{X_1, X_2, \dots, X_N} \left[\sum_{i=1}^N a_i X_i \right] = \sum_{i=1}^N a_i E_{X_i} [X_i]$$

$$a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{bmatrix}$$

$$E_X [a^T X] = a^T E_X [X]$$

Who remembers formula for
sum of variances?

formula for sum of variances

$$\text{var}(X_1 + X_2) = \text{var}(X_1) + \text{var}(X_2) + 2 \text{cov}(X_1, X_2)$$

$$\text{var}\left(\sum_{i=1}^N X_i\right) = \sum_{i=1}^N \text{var}(X_i) + \sum_{i=1}^N \sum_{\substack{j=1 \\ i \neq j}}^N \text{cov}(X_i, X_j)$$

$$N=2$$

$$C_X = \begin{bmatrix} \text{var}(X_1) & \text{cov}(X_1, X_2) \\ \text{cov}(X_2, X_1) & \text{var}(X_2) \end{bmatrix}$$

$$\text{var}(X_1 + X_2) = [1 \ 1] C_X \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Covariance matrix

$$C_X = \begin{bmatrix} \text{var}(X_1) & \text{cov}(X_1, X_2) & \dots & \text{cov}(X_1, X_N) \\ \text{cov}(X_2, X_1) & \text{var}(X_2) & \dots & \text{cov}(X_2, X_N) \\ \vdots & & & \vdots \\ \text{cov}(X_N, X_1) & \text{cov}(X_N, X_2) & \dots & \text{var}(X_N) \end{bmatrix}$$

Properties of covariance matrix

1. Covariance matrix is symmetric
2. Covariance matrix for uncorrelated random variables is a diagonal matrix

3. Covariance matrix is positive semidefinite

$$a^T C_X a \geq 0 \quad \text{for all vectors } a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

4. Covariance matrix of $Y=AX$ is easily determined:

$$C_Y = A C_X A^T$$

5. Covariance matrix can be always diagonalized. (by transforming a random vector to a new vector such that the corresponding covariance matrix is diagonal, we can *decorrelate* the random variables)

- **Example.** A computer system receives messages over three communication lines. Let X_i be the number of messages received on line i in one hour. The joint pmf is given by

$$p(x_1, x_2, x_3) = (1 - a_1)(1 - a_2)(1 - a_3)a_1^{x_1}a_2^{x_2}a_3^{x_3}$$

Find individual pmfs

Example 3 communication lines

$$\begin{aligned} P_{X_3}(x_3) &= \sum_{i=1}^{\infty} \sum_{k=1}^{\infty} (1-a_1)(1-a_2)(1-a_3) a_1^i a_2^k a_3^{x_3} = \\ &= (1-a_1)(1-a_2)(1-a_3) a_3^{x_3} \underbrace{\sum_{i=1}^{\infty} a_1^i}_{\frac{1}{1-a_1}} \underbrace{\sum_{k=1}^{\infty} a_2^k}_{\frac{1}{1-a_2}} = \\ &= (1-a_3) a_3^{x_3} \end{aligned}$$

- Q: are lines independent?

Continuous N-dimensional random variables

- N-dimensional joint pdf

$$p_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N)$$

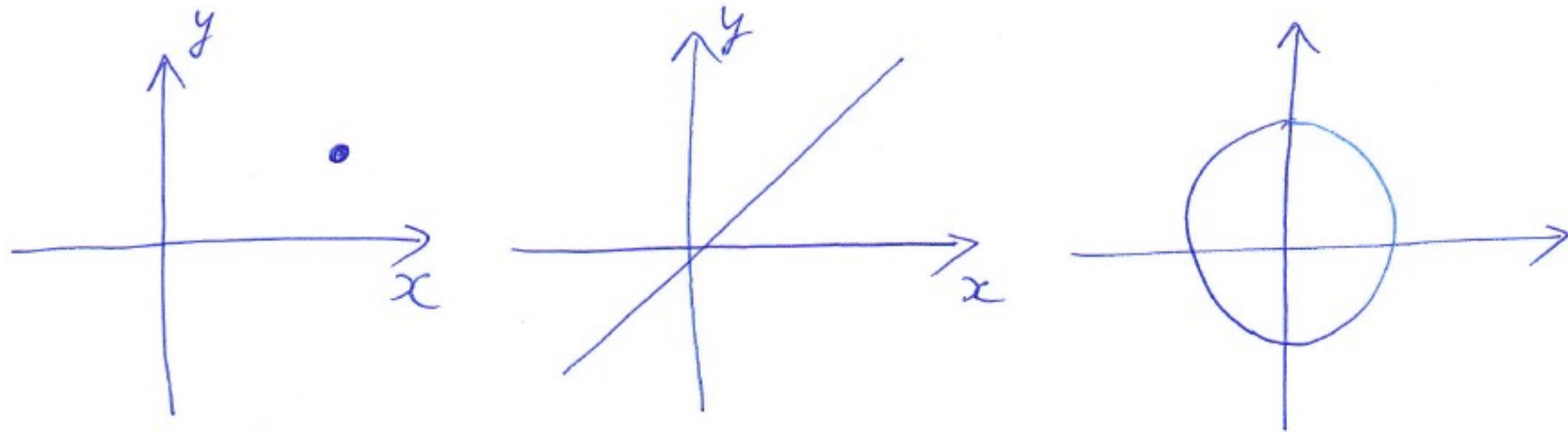
- The following properties should be valid:

$$p_X(x) \geq 0$$
$$\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} p_{X_1 \dots X_N}(x_1, \dots, x_N) dx_1 \dots dx_N = 1$$

- Probability of an event A is given by

$$P[A] = \int \dots \int_A p_{X_1 \dots X_N}(x_1, \dots, x_N) dx_1 \dots dx_N$$

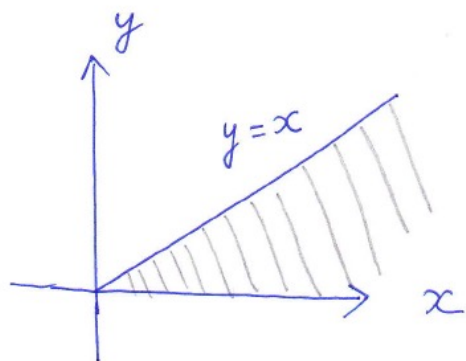
Zero probability event in two dimensions



Example: covariance

joint pdf for X and Y :

$$f_{X,Y}(x,y) = \begin{cases} ce^{-x}e^{-y}, & 0 \leq y < x < \infty \\ 0, & \text{otherwise} \end{cases}$$



The constant c is found from the normalization condition:

$$1 = \int_0^{+\infty} \int_0^x ce^{-x}e^{-y} dy dx = \frac{c}{2} \quad \Rightarrow \quad c=2$$

Example: covariance (ctnd)

The marginal pdfs:

$$f_X(x) = \int_0^x 2e^{-x}e^{-y} dy = 2e^{-x}(1-e^{-x}), \quad x \in [0, \infty)$$

$$f_Y(y) = \int_0^\infty 2e^{-x}e^{-y} dx = 2e^{-2y}, \quad y \in [0, \infty)$$

$$E[X] = \frac{3}{2}, \quad \text{Var}(X) = \frac{5}{4}$$

$$E[Y] = \frac{1}{2}, \quad \text{Var}(Y) = \frac{1}{4}$$

Example: covariance (ctnd)

$$\begin{aligned} E[XY] &= \int_0^{\infty} \int_0^x xy \cdot 2e^{-x}e^{-y} dy dx = \\ &= \int_0^{\infty} 2xe^{-x}(1 - e^{-x} - xe^{-x}) dx = 1 \end{aligned}$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 1 - \frac{3}{4} = \frac{1}{4}$$

$$\text{Corr}(X, Y) = \frac{\frac{1}{4}}{\sqrt{\frac{5}{4}} \sqrt{\frac{1}{4}}} = \frac{1}{\sqrt{5}}$$

Example – multi-component system

A communication system consists of n components. Each of components independently functions with probability p . The system is able to operate if at least half of its components functions. As an engineer, we should choose if we should construct a 5-component system or a 3-component system. In other words, for which values of p is a 5-component system more likely to operate effectively than a 3-component system?

For 5-component system:

$$\begin{aligned} P_1 &= P\{X \geq 3\} = P\{X=3\} + P\{X=4\} + P\{X=5\} = \\ &= 1 - P\{X < 3\} = 1 - P\{X=0\} - P\{X=1\} - P\{X=2\} = \\ &= \binom{5}{3} p^3 (1-p)^2 + \binom{5}{4} p^4 (1-p) + \binom{5}{5} p^5 (1-p)^0 \end{aligned}$$

For 3-component system:

$$\begin{aligned} P_2 &= P\{Y \geq 2\} = P\{Y=2\} + P\{Y=3\} = \\ &= \binom{3}{2} p^2(1-p) + \binom{3}{3} p^3(1-p)^0 \end{aligned}$$

We would like to choose p such that $P_1 \geq P_2$

$$\Rightarrow 10p^3(1-p)^2 + 5p^4(1-p) + p^5 \geq 3p^2(1-p) + p^3$$

$$3(1-p)^2(2p-1) \geq 0$$

$$p \geq \frac{1}{2}$$

Example – error control by retransmissions

A sends a message to *B*. At *B* an error detection mechanism is applied and upon detection an error, retransmission is required. Let $p = 0.9$ is probability to receive message correctly.

- (a) What is the probability that the message will be transmitted more than twice?
- (b) How many times on average the message will be transmitted?

Let M be a random variable denoting a number of times a certain message has been transmitted. $M = \{1, 2, \dots\}$. We derive a general formula for probability that more than k transmissions is required to receive the message:

$$P(M > k) = P(M = k+1) + P(M = k+2) + \dots = (1-p)^k p + (1-p)^{k+1} p + \dots =$$

$$p(1-p)^k \sum_{i=0}^{\infty} (1-p)^i = (1-p)^k$$

- (a) Thus,

$$P(M > 2) = (1 - 0.9)^2 = 10^{-2}$$

- (b) M is a geometric random variable.

$$E[M] = 1/p \approx 1.11$$

